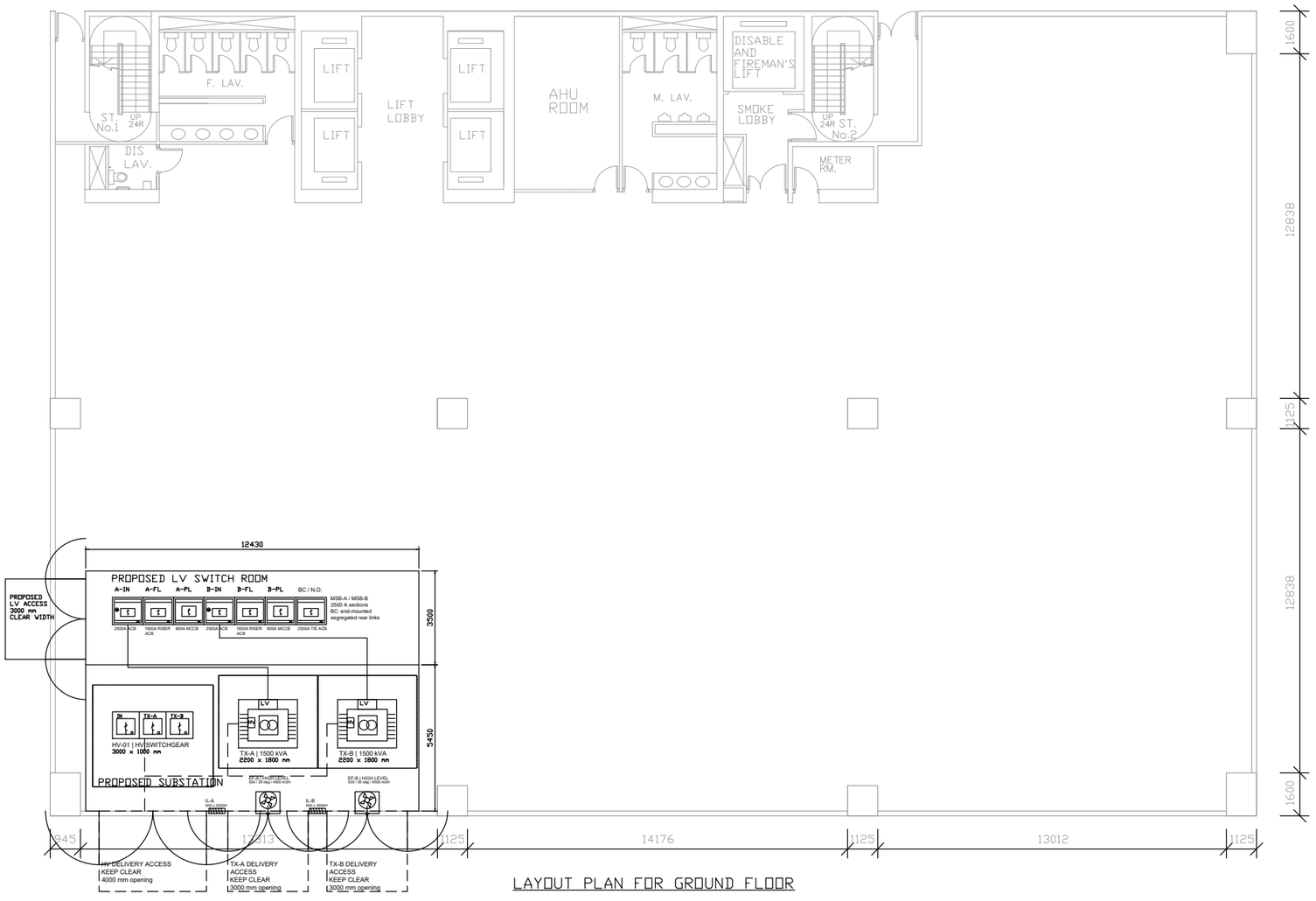
# TASK 1 | GROUND-FLOOR PLANTROOM LOCATION

A3 LANDSCAPE | SCALE 1:200 | DIMENSIONS IN mm | GROUND FLOOR




LAYOUT PLAN FOR GROUND FLOOR

PROPOSED LOCATION
Substation: 12430 x 5450 mm.
Adjacent LV room: 12430 x 3500 mm (proposed).
See 1:75 plantroom detail.

LOCATION REASONS
External plant access.
Short transformer-to-incomer interfaces.
Separation from lift/stair/AHU core.
Independent west entrance to LV room.

LOAD ASSESSMENT
Offices: 1644.800 kVA.
AC: 1000 kW cooling -> 526.039 kW electrical -> 584.488 kVA at assumed PF 0.90.
Public services: 200 kVA aggregate allowance.
Two-zone conservative envelope: 2672.216 kVA.
Propose 2 x 1500 kVA, with equal zone allocation; no full-load N-1.

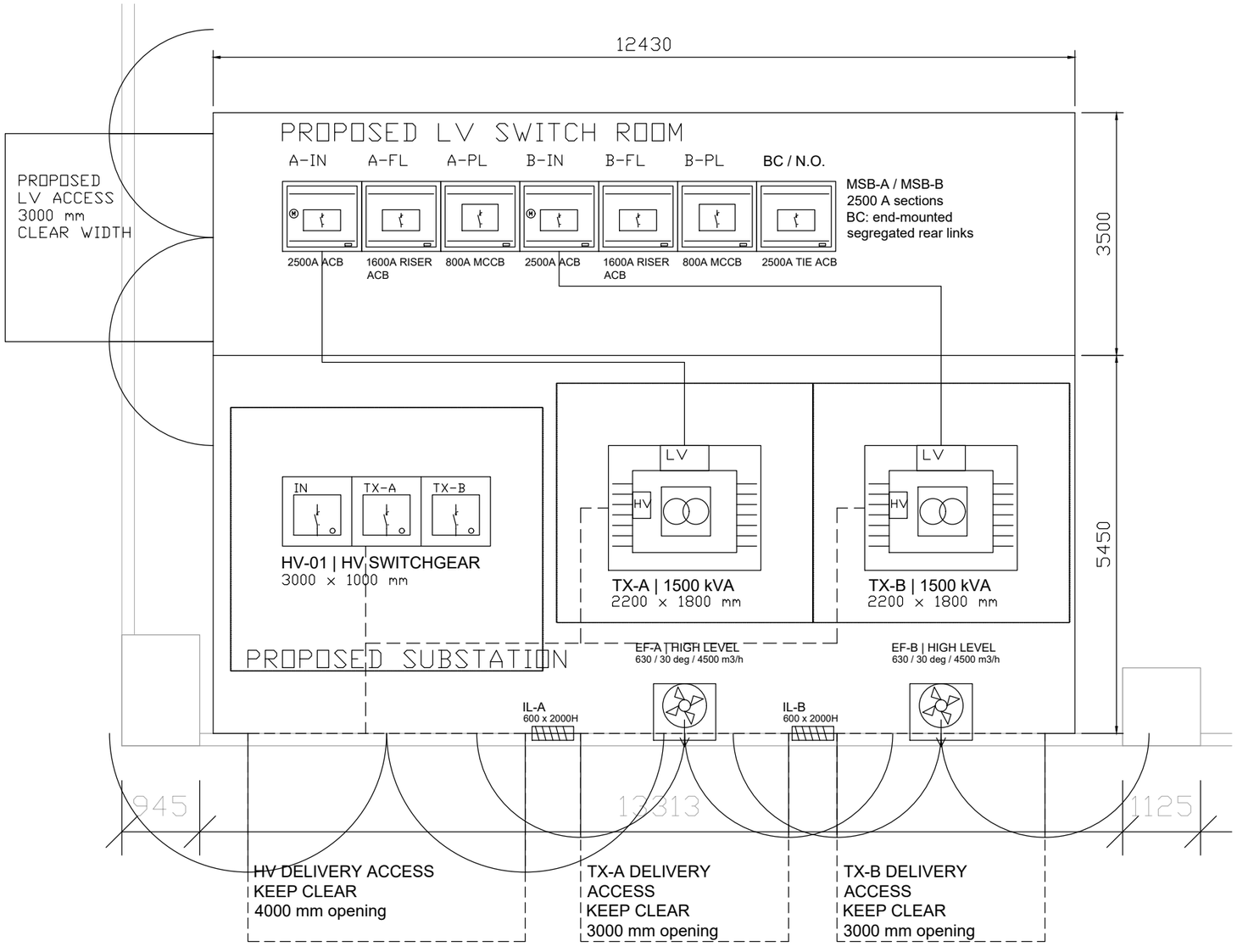
SITE COORDINATION
Confirm external access, >=3 m clear delivery route and unloading within 20 m.
Coordinate openings, structure, flood/fire provisions and supplier equipment dimensions.

GROUP 1 | TASK 1

Design basis: CLP COP215 Rev07; CLP COP101 v16.

# TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

A3 LANDSCAPE | SCALE 1:75 | DIMENSIONS IN mm | PLANTROOM DETAIL




LINE KEY: dashed line = HV trench; solid line = LV route; dashed box = delivery clearance. Equipment details diagrammatic.

LV INTERFACES / HV ROUTES

LV trench: 1000 W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.

HV centre-lines: 1000 W at panels; 600 W branches; final 300 W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3

EQUIPMENT

TX-A/B: CLP 1500 kVA each; demand 2672.216 kVA.

HV-01: incomer + two transformer protection panels.

MSB: two 2500 A sections, seven 1140 mm bays.

BC: normally-open 4P end coupler with segregated rear links. See main schematic for interlocking.

CLEARANCES / BUILDING WORKS

HV: rear 1000, front 1800, sides 750 mm.

TX: LV terminals 900, other sides 750 mm.

LV board: front 1500, rear 990 mm; total width 7980 mm.

Proposed 2 h FRR partitions, sealed penetrations and FFL 150 mm above external level.

ACCESS AND DOORS

Proposed south delivery access: >=3000 mm clear route; unloading within 20 m.

HV opening 4000 mm; TX openings 3000 mm each; proposed height 2800 mm.

Outward-opening doors shown at 90 degrees; adjacent doors used sequentially.

TX-A wicket access; independent west LV entrance.

VENTILATION

EF-A/B: 630 mm / 30 deg / 700 rpm; each 4500 m3/h at 70 Pa; <=58 dB(A) at 1 m (COP101 6.6).

IL-A/B: 600 x 2000H mm, >=50% free area; total 1.20 m2.

High-level ducts: 900 x 750 mm; discharge >=2.5 m above street. Thermal calculation: Task 1.

CLP COP215 Rev07; COP101 v16, 4.3 and 6.1-6.7.

Equipment selections and calculations accompany this drawing.

# TASK 1 | MAIN DISTRIBUTION, PROTECTION AND METERING

REV04 | 380/220 V, 3-phase 4-wire + PE | Single-line diagram, not to scale | Normal supply only

## TX-A | CLP 1500 kVA

Utility HV protection
11 kV / 380 V
uk = 6% assumed

ACB-A | 4P draw-out
MTZ2 25H1 / 6.0X LSIG
2500AF; Ir 2250 A; 66 kA

4-pole sensors > LSIG > trip
Isd 4500 A; tsd 0 s; I2t OFF
Ig 500 A; tg 0.2 s, I2t OFF

MSB-A | 2500 A section

BC | 4P, N.O.
2500AF / 2250AT
MTZ2 25H1, 66 kA

A-FL | 1600AF
Ir 1440 A; 66 kA
KTC2500 busway

A-PL | 800/720 A
4P, 70 kA
Plant / public

F1 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F2 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F3 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F4 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F5 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

TPN MCCB BEFORE each tenant meter:
O1/O2 100 A; O3 125 A + CT meter.
F1/O1 63 A; all tenant MCCBs70 kA.

## TX-B | CLP 1500 kVA

Utility HV protection
11 kV / 380 V
uk = 6% assumed

ACB-B | 4P draw-out
MTZ2 25H1 / 6.0X LSIG
2500AF; Ir 2250 A; 66 kA

4-pole sensors > LSIG > trip
Isd 4500 A; tsd 0 s; I2t OFF
Ig 500 A; tg 0.2 s, I2t OFF

MSB-B | 2500 A section

B-FL | 1600AF
Ir 1440 A; 66 kA
KTC2500 busway

B-PL | 800/720 A
4P, 70 kA
Plant / public

F6 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F7 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F8 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F9 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F10 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

TPN MCCB BEFORE each tenant meter:
O1/O2 100 A; O3 125 A + CT meter.
F1/O1 63 A; all tenant MCCBs70 kA.

## F1 / OFFICE 1 | DETAILED FEEDER

From F1 320 A floor MCCB
MS-O1: ComPacT NSX100H
C10H4TM063, 4P4D TM63D
Ir 63 A; Icu/Ics 70 kA
Suitable for isolation

M CLP tenant meter
Immediately after MS-O1

4C35 mm2 PVC Cu
+ separate CPC16 mm2
L = 31.5119 m
28.5119 plan + 3 m allowance

DB-O1: 100 A 4P isolator
15 x 1P+N Type A 30 mA RCBO
7 lighting + 8 power circuits

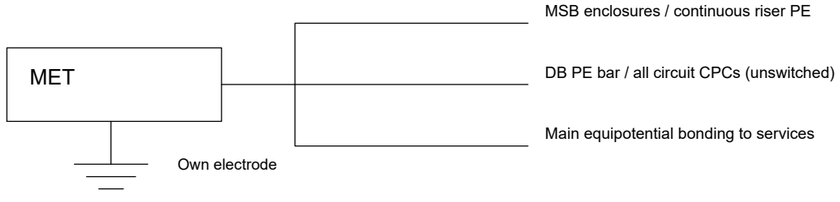
PE continuous; neutral switched with phases.
No customer N-PE link at DB.

## ACB-A / ACB-B / BC INTERLOCK

Electrical AND mechanical interlock
Normal: CLOSED / CLOSED / OPEN
Transfer from A: CLOSED / OPEN / CLOSED
Transfer from B: OPEN / CLOSED / CLOSED
ALL THREE CLOSED: PROHIBITED

Open/isolate failed incomer before closing BC.
Before transfer, shed load to <=1480.9 kVA
(2250 A at 380 V); no full-load N-1.
Utility return: open BC before restoring incomer.

## CUSTOMER EARTHING




## BASIS AND FAULT DUTY

CLP GuideToSupply v10: sections 2, 3, 7.4, 11.
MSB PFC <=56.92 kA incl. transfer motors.
Own MET/electrode; any CLP bond only by agreement, >=70 mm2 Cu. See Task 3 for loop, disconnection and conductor checks.
CT meters: CLP ratios +16 A HRC voltage fuses; Guide sections9.11 and9.16.

2672.216 kVA building demand | 1336.108 kVA / 2030 A per normal section | CT/sensor arrangement subject to CLP acceptance